MACHINE LEARNING-BASED DETECTION OF LUMBAR SPINE DEGENERATIVE DISEASES USING MRI IMAGES

DANIEL INNOCENT KINYUNYU

A PROJECT REPORT SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF BACHELOR OF ENGINEERING IN DATA SCIENCE AND TECHNOLOGY OF MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY

AUGUST, 2026

MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY (MUST)



COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY

COMPUTER SCIENCE AND ENGINEERING

BACHELOR OF ENGINEERING IN DATA SCIENCE

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22100533590051

UE/BEDS/25/19855

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ABSTRACT

Lumbar spine degeneration is among the leading causes of low back pain and disability worldwide, yet radiological grading of its severity is time-consuming and subject to inter-reader variability, a challenge that is more acute in resource-constrained settings that lack specialist radiologists. This project developed Spinova, a deep learning based Clinical Decision Support System (CDSS) for automated classification of lumbar spine degenerative conditions from magnetic resonance imaging (MRI). The system adopts a two-stage localize-then-classify design: level/condition coordinates are used to crop the region of interest before classification, rather than feeding whole slices to the network, addressing a known weakness of earlier whole-image approaches.

An EfficientNet-B4 backbone enhanced with Coordinate Attention and a Spatial Attention Module was fine-tuned to classify each crop into Normal/Mild, Moderate, or Severe categories, using class-weighted loss, minority-class oversampling, and augmentation to counter severe class imbalance. An out-of-distribution "MRI Gatekeeper" module, built on a frozen MobileNetV3-Small embedding backbone, was developed to reject non-lumbar-spine images before they reach the severity models. For explainability, SHAP attribution maps and Monte Carlo Dropout uncertainty estimates were integrated alongside Grad-CAM visualisation. To meet deployment constraints on modest hospital hardware, the trained model was distilled into a lightweight MobileNetV3-Small student and exported to ONNX with INT8 quantisation for CPU inference.

On the held-out RSNA 2024 Lumbar Spine test set, the hybrid model achieved 79.15% crop-level accuracy, a macro-F1 score of 0.594, an AUC-ROC (macro) of 0.853, and a Cohen's Kappa of 0.415, with the distilled, quantised student retaining comparable accuracy (77.31%) while running a simulated full study (25 crops) in under half a second. A Django REST Framework backend and a React-based frontend were built to expose the model through an interactive clinical workspace supporting image upload, point-based level/condition marking, and visual result review. The findings demonstrate that a localized, attention-guided, explainable, and edge-deployable pipeline is feasible for lumbar spine screening support, while also identifying gaps including automatic landmark localization and 3-D segmentation-constrained explanations that remain necessary before clinical deployment.

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ACKNOWLEDGEMENT

I am grateful to the Almighty God for the strength, wisdom, and perseverance granted throughout this study. I sincerely thank my supervisor, Dr. Sadiki Lameck Kusyama, for his guidance, patience, and constructive criticism during the design, development, and write-up of this project. I also extend my appreciation to the Department of Data Science and Technology at Mbeya University of Science and Technology for the academic support offered throughout the Bachelor of Engineering in Data Science and Technology programme.

Special thanks go to my colleagues and family for their encouragement, and to everyone whose feedback shaped the direction of the Spinova system

DECLARATION

I, Daniel Innocent Kinyunyu, declare that this project report is my own original work and that it has not been presented and will not be presented to any other Higher Learning Institution for a similar or any other academic award.

Signature: _____ Date: _____

CERTIFICATION

The undersigned certifies that he has read and hereby recommends for acceptance by Mbeya University of Science and Technology a project report titled “Spinova: A Deep Learning-Based Clinical Decision Support System for Lumbar Spine Degenerative Disease Classification with Explainable Artificial Intelligence” in partial fulfilment of the requirements for the award of the Degree of Bachelor of Engineering in Data Science and Technology of Mbeya University of Science and Technology.

_____

Dr. Sadiki Lameck Kusyama (Supervisor's Name)

_____

Supervisor's Signature

Date: _____

DEDICATION

This project report is dedicated to my family, for their unwavering love, patience, and encouragement throughout my studies, and to the Almighty God, for the strength, wisdom, and perseverance granted throughout this journey.

ABBREVIATIONS AND ACRONYMS

AI	Artificial Intelligence
API	Application Programming Interface
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
CA	Coordinate Attention
CDSS	Clinical Decision Support System
CNN	Convolutional Neural Network
CPU	Central Processing Unit
CRISP-DM	Cross-Industry Standard Process for Data Mining
DICOM	Digital Imaging and Communications in Medicine
ERD	Entity-Relationship Diagram
Grad-CAM	Gradient-weighted Class Activation Mapping
GPU	Graphics Processing Unit
JSON	JavaScript Object Notation
JWT	JSON Web Token
MC-Dropout	Monte Carlo Dropout
MRI	Magnetic Resonance Imaging
MUST	Mbeya University of Science and Technology
ONNX	Open Neural Network Exchange
OOD	Out-of-Distribution
REST	Representational State Transfer
RSNA	Radiological Society of North America
SAM	Spatial Attention Module
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
TTA	Test-Time Augmentation
UI	User Interface
UML	Unified Modeling Language

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CHAPTER ONE

INTRODUCTION

1.1 Background

Low back pain associated with lumbar spine degeneration spinal canal stenosis, neural foraminal narrowing, and subarticular stenosis is one of the leading causes of disability globally. Diagnosis relies on magnetic resonance imaging (MRI) interpreted by radiologists, who grade severity per spinal level and condition. This grading process is time-consuming, requires specialist expertise, and is subject to inter-reader variability, a challenge that is compounded in low-resource settings, including many parts of Tanzania and the wider East African region, where access to specialist radiologists is limited.

Advances in deep learning, particularly convolutional neural networks (CNNs), have shown strong performance in medical image classification tasks. Public benchmark datasets such as the Radiological Society of North America (RSNA) 2024 Lumbar Spine Degenerative Classification dataset have enabled research into automated grading of stenosis severity. However, published work on this dataset shows that approaches which feed whole MRI slices into a network, or that collapse multiple condition-level labels into a single severity score, under-perform because the pathological region occupies only a small fraction of each slice.

This project responds to that gap by building Spinova, a two-stage, localize-then-classify deep learning pipeline integrated into a full-stack Clinical Decision Support System (CDSS), so that the resulting model is not only accurate but also explainable, lightweight enough for modest hospital hardware, and usable through a web-based clinical workspace.

1.2 Problem Statement

Manual grading of lumbar spine MRI studies is slow and dependent on the availability of experienced radiologists, a resource that is scarce in many Tanzanian and regional healthcare facilities. Prior automated approaches to this classification task that operate on whole MRI slices, or that merge distinct condition-level severities into a single output, lose the localized signal that distinguishes Normal/Mild, Moderate, and Severe degeneration, and tend to under-perform, particularly on the clinically important Moderate and Severe classes, which are also the least represented in typical datasets. In addition, most published deep learning solutions for this task are evaluated purely on accuracy metrics, without addressing three requirements that matter for real clinical use: (i) explainability of the model's predictions, (ii) robustness to inputs that are not lumbar spine MRIs at all, and (iii) the ability to run inference on affordable, CPU-

only hardware typical of under-resourced clinics. This project therefore investigates whether a localized, attention-guided classification pipeline, combined with explainability, out-of-distribution rejection, and edge-optimised deployment, can address these gaps within a usable clinical decision support system.

1.3 Objectives

1.3.1 Main Objective

To develop a machine learning-based detection system for lumbar spine degenerative diseases using MRI images for accurate multi-class classification through incorporating explainable AI capabilities.

1.3.2 Specific Objectives

i. To acquire and preprocess a multimodal dataset comprising lumbar spine MRI images and associated clinical variables.

ii. To train and evaluate an attention-augmented CNN model for localized, multi-class lumbar pathology severity classification.

iii. To design and implement a user-interface, web-based system that integrates the trained model for real-time classification and clinical decision support.

1.4 Project Significance

The project contributes a reference pipeline for localized medical image classification that other students and researchers at MUST can build on for similar imaging tasks. Practically, a working, explainable, and lightweight CDSS prototype demonstrates how AI-assisted screening tools can be designed for resource-constrained clinical settings such as those found in Mbeya and other regions of Tanzania, potentially reducing the diagnostic burden on limited radiology staff. Academically, the project fulfils the requirement for the Bachelor of Engineering in Data Science and Technology dissertation and demonstrates the application of data science and deep learning techniques to a real health-sector problem.

1.5 Scope and Limitations of the Project

The project covers the design, training, and evaluation of the deep learning classification model on the publicly available RSNA 2024 Lumbar Spine Degenerative Classification dataset, the development of explainability and out-of-distribution rejection modules, model compression for edge deployment, and the development of a Django/React web application that exposes the model through a clinical workspace interface.

The project does not cover automatic landmark/level localisation on unlabelled clinical scans, 3-D segmentation-constrained explanation, prospective clinical validation, or regulatory certification of the system; these are identified as areas for further work in Chapter Five. The system is also limited by its reliance on a single public dataset for training, meaning performance on locally-sourced Tanzanian MRI scans, which may differ in scanner type and imaging protocol, has not been independently verified.

CHAPTER TWO

LITERATURE REVIEW

This chapter reviews existing systems developed for automated classification of lumbar spine MRI studies, drawing out their design choices, strengths, and limitations, before presenting the system proposed in this project and how it addresses the gaps identified.

2.1 Existing System

Automated lumbar spine degeneration classification systems reported in the literature broadly fall into two categories, distinguished chiefly by whether the classifier operates on the whole MRI slice or on a localized region of interest.

2.1.1 System 1: Whole-Slice, Single-Label Classification Systems

The first category of systems feeds entire MRI slices directly into a convolutional neural network and collapses the several condition-level severities present in a study spinal canal stenosis, neural foraminal narrowing, and subarticular stenosis, each assessed at five vertebral levels into a single severity label per slice or per study. These systems are comparatively simple to build and train, since they require no coordinate-level annotation and can reuse standard image-classification pipelines with minimal modification.

However, this simplicity comes at a cost. Because the pathological region of interest occupies only a small fraction of each slice, the network must learn to ignore a large amount of irrelevant background, which dilutes its attention and lowers accuracy, particularly for the clinically important Moderate and Severe classes. Collapsing multiple, anatomically distinct conditions into a single label also destroys information that a clinician would need for a level-by-level, condition-by-condition report, limiting the clinical usefulness of the output even when overall accuracy appears acceptable.

2.1.2 System 2: Two-Stage, Localize-then-Classify, Attention-Augmented Systems

The second category, reported more recently in work built around the RSNA 2024 Lumbar Spine dataset, first localizes the disc level or region of interest and then classifies a crop centred on that location, in some cases augmented with attention mechanisms that let the network weight the most diagnostically relevant channels or spatial regions of the cropped feature map. These two-stage, attention-augmented pipelines have been reported to substantially outperform whole-slice baselines, with area-under-the-curve figures around 0.97 for canal stenosis detection, confirming that localisation combined with attention is an effective design choice for this classification task.

Even so, published two-stage systems in this category are typically evaluated purely as classification models. They generally do not integrate explainability outputs (such as SHAP attribution or predictive uncertainty) alongside the prediction, do not include a safeguard against out-of-distribution inputs such as non-lumbar-spine images, and are not compressed or packaged for CPU-only deployment on affordable hospital hardware. They remain, in other words, research-grade classification models rather than deployable, end-to-end clinical decision support systems.

2.2 Proposed System

Spinova, the system proposed and developed in this project, builds on the localize-then-classify, attention-augmented design of System 2 using an EfficientNet-B4 backbone enhanced with a Coordinate Attention block and a Spatial Attention Module while directly closing the integration gap identified in both existing systems. Specifically, Spinova adds: (i) an MRI Gatekeeper module, built on a frozen MobileNetV3-Small embedding backbone, that rejects non-lumbar-spine or otherwise out-of-distribution images before they reach the severity classifier; (ii) a multi-method explainability suite combining SHAP attribution maps, Monte Carlo Dropout predictive uncertainty, and Grad-CAM visualisation, rather than a bare class label; (iii) knowledge distillation of the trained model into a lightweight MobileNetV3-Small student, followed by INT8 post-training quantisation and export to the ONNX format, so that the model can run in well under a second per study on CPU-only hospital hardware; and (iv) a complete, working Clinical Decision Support System a Django REST Framework backend and a React-based frontend through which a clinician can upload a study, mark levels and conditions, and receive live predictions, confidence, uncertainty, and a surgical risk-tier flag.

The conceptual framework of Spinova links four stages: (1) data preparation cropping labelled MRI regions of interest and correcting class imbalance through oversampling, class-weighted loss, and augmentation; (2) model development training the attention-augmented backbone alongside the MRI Gatekeeper; (3) explainability and compression attaching SHAP, MC-Dropout, and Grad-CAM outputs, then distilling and quantising the model; and (4) deployment exposing the compressed model through the REST API and clinical workspace. Each stage's output becomes the input of the next, forming a single pipeline from raw MRI data to a clinician-facing decision support interface, and it is this end-to-end integration rather than the classification architecture alone that distinguishes Spinova from the two categories of existing systems reviewed above.

CHAPTER THREE

METHODOLOGY

3.1 Introduction

This chapter describes how the Spinova project was carried out: the approach and design adopted, how data was collected and analysed, and how the classification model and the surrounding clinical decision support system were developed. The project adopted a quantitative, experimental, design-science approach, in which an artefact the Spinova classification model and CDSS was iteratively designed, built, and evaluated against quantitative performance metrics, following the Cross-Industry Standard Process for Data Mining (CRISP-DM) methodology (Chapman et al., 2000). Development proceeded in versioned phases, each building on lessons from the previous version, including an earlier whole-slice, multi-label design that was replaced after diagnosing label-image mismatches and under-performance.

3.2 Data Collection

The primary dataset used was the publicly available RSNA 2024 Lumbar Spine Degenerative Classification dataset, released by the Radiological Society of North America, comprising lumbar spine MRI studies with per-level, per-condition severity labels and coordinate annotations (train_label_coordinates.csv) marking the pixel location of each condition at each vertebral level. The data used were secondary, pre-existing DICOM-format MRI images and structured severity/coordinate labels; no primary data collection from human subjects was undertaken by the researcher.

The population comprised all labelled coordinate rows in the dataset, each linked to a specific DICOM slice, level, and condition. After resolving each coordinate to its DICOM image and label and dropping rows with missing labels or files, the resulting crop-level dataset was split at the study level (not the slice level, to avoid data leakage across splits) using stratified group splitting. The final split produced 29,269 crops (1,186 studies, 60.2%) for training, 9,746 crops (396 studies, 20.0%) for validation, and 9,642 crops (392 studies, 19.8%) for testing. StratifiedGroupKFold was used to allocate whole studies (not individual crops) to these folds, stratified by severity class and grouped by study identifier, ensuring that no study's crops appeared in more than one split. Within the training split, class imbalance was further addressed using a WeightedRandomSampler, which oversamples minority-class crops in proportion to the inverse of their class frequency, serving as an image-domain equivalent of SMOTE.

The project used a de-identified, publicly released research dataset made available under the competition's data-use terms, and did not involve direct interaction with human participants. The prototype system is explicitly positioned as a decision-support aid rather than an autonomous diagnostic tool; risk-tier thresholds derived from model outputs are flagged in the system as drafts requiring clinical review before any real report is issued, and the system's out-of-distribution gatekeeper is intended to reduce the risk of the model producing a confident but meaningless output on a non-lumbar-spine image.

3.3 Data Analysis

For each labelled coordinate, a square region was cropped in the original DICOM pixel space around the labelled point (with crop size a fraction of the image's shorter dimension) and resized to the model's input resolution, keeping the lesion centred and scale-consistent across scanners. This preprocessing step converted the raw, whole-slice DICOM data into the well-posed, single-label crops used for training and evaluation.

Model performance was analysed using accuracy, macro-averaged precision, recall (sensitivity), F1-score, macroAUC-ROC, Cohen's Kappa, and a competition-style weighted log loss, computed at the crop level on the held-out test set and reported per class through a classification report and confusion matrix. Test-Time Augmentation (five views) was additionally evaluated for its effect on the primary hybrid model, and the effect of INT8 quantisation on the distilled student's accuracy and inference latency was measured through a simulated full-study (25-crop) inference benchmark.

Validity was addressed by resolving an earlier label-image mismatch (where slices had been used for labels regardless of the true imaging plane) and by using study-level, leakage-checked splits so that no patient study contributed crops to more than one split. Reliability of evaluation was addressed by reporting a standard set of metrics on a held-out test set that was not used for training or threshold tuning, with post-hoc decision-threshold tuning performed only on the validation set.

3.4 System Development

The classification backbone was an EfficientNet-B4 network pretrained on ImageNet and fine-tuned with progressive layer unfreezing, augmented with a Coordinate Attention block and a Spatial Attention Module inserted after feature extraction. Training used a class-weighted categorical cross-entropy loss (weighting Normal/Mild = 1, Moderate = 2, Severe = 4, mirroring

the competition's severity weighting), a differential learning-rate AdamW optimiser, and a warm-up plus cosine learning-rate schedule.

An MRI Gatekeeper module was developed separately as an out-of-distribution detector: a frozen, ImageNet-pretrained MobileNetV3-Small extracts embeddings from known-good lumbar spine MRI slices to build a reference distribution, and any new image whose embedding falls far outside that distribution is rejected before reaching the severity models. For explainability, SHAP attribution maps and Monte Carlo Dropout uncertainty estimates were computed for each prediction, complementing Grad-CAM.

For deployment, the trained model was distilled into a MobileNetV3-Small student (same 224×224-crop input, three-class severity output); the student was quantised to INT8 for CPU inference; and both the student model and the gatekeeper backbone were exported to the ONNX format together with a model_config.json describing preprocessing constants (crop fraction, image size, normalisation statistics) and class/condition names, so that the Django backend could load them without requiring a full PyTorch installation.

These artefacts were integrated into a working system: a Django REST Framework backend (with endpoints for study upload and prediction) loads the ONNX models via ONNX Runtime and reads configuration from the exported JSON files rather than hardcoded constants, while a React (Vite, TypeScript, Tailwind CSS) frontend provides a marketing site (Home, About, How It Works) and an authenticated Workspace page where a clinician can upload an MRI image, mark points for the levels and conditions to be screened, and receive live severity, confidence, and risk-tier predictions returned by the trained model.

The model was developed in Python using PyTorch and torchvision (EfficientNet-B4, MobileNetV3-Small), scikit-learn (StratifiedGroupKFold and evaluation metrics), pydicom (DICOM reading), and Optuna for hyperparameter search, on a Kaggle T4×2 GPU environment. The clinical decision support system was implemented with a Django REST Framework backend (serving predictions through an ONNX Runtime inference endpoint) and a React (Vite, TypeScript, Tailwind CSS) frontend.

3.5 Pipeline and System Workflow Summary

This section brings together the stages described above into a single pipeline view and a stage-by-stage workflow summary, from the acquisition of the RSNA 2024 dataset through to the deployment of the distilled, explainable model within the Spinova clinical decision support system.

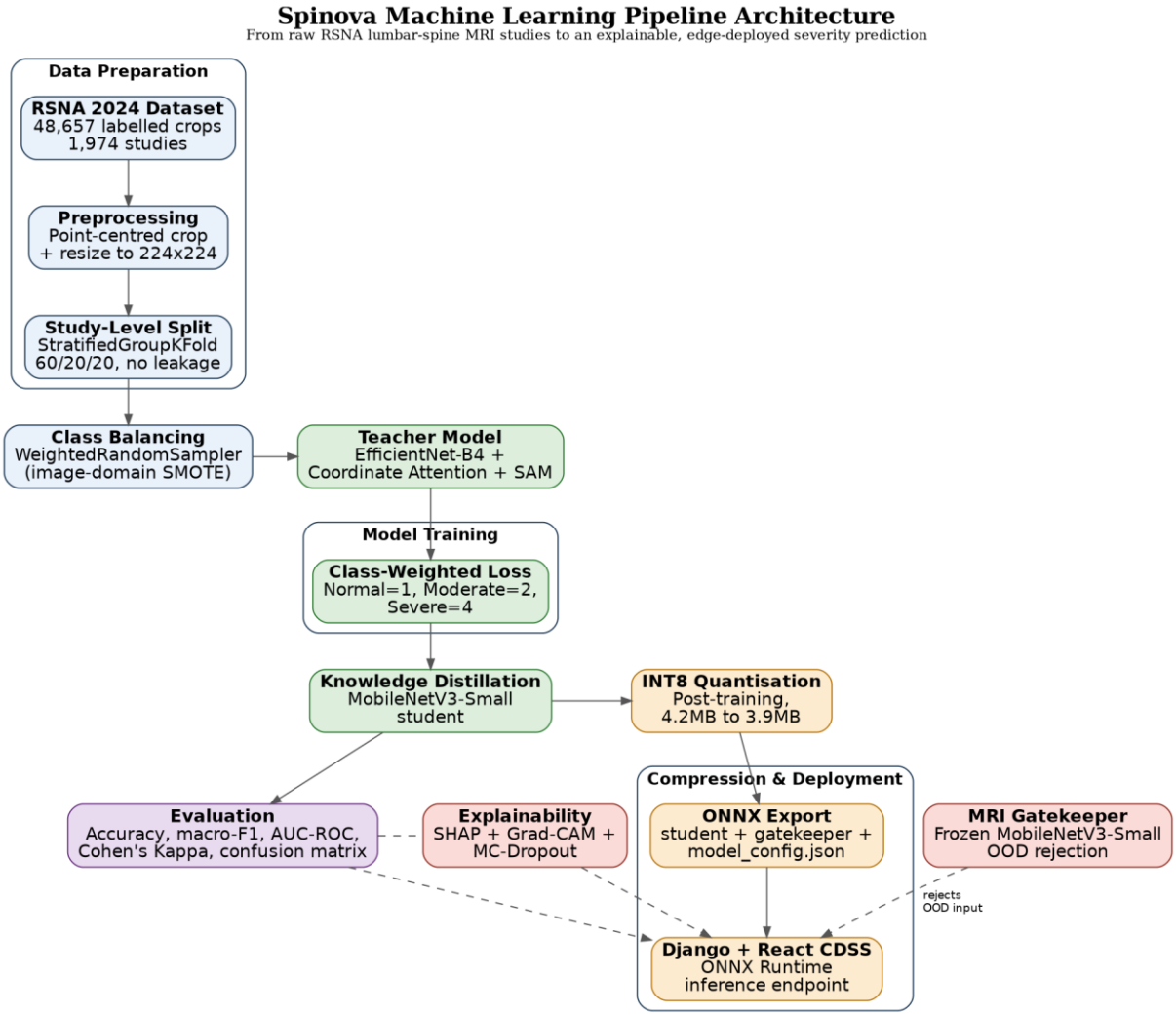


Figure 1: Machine Learning Pipeline Architecture – Spinova CDSS

Figure 1 shows the overall pipeline architecture. Data preparation (dataset ingestion, point-centred cropping, and leakage-safe study-level splitting) feeds a class-balanced training set to the EfficientNet-B4 teacher model, which is then compressed through knowledge distillation into a MobileNetV3-Small student and exported with INT8 quantisation for CPU inference. The MRI Gatekeeper and the SHAP/Grad-CAM/MC-Dropout explainability layer operate alongside the trained model to support safe, interpretable predictions before results reach the Django and React clinical workspace.

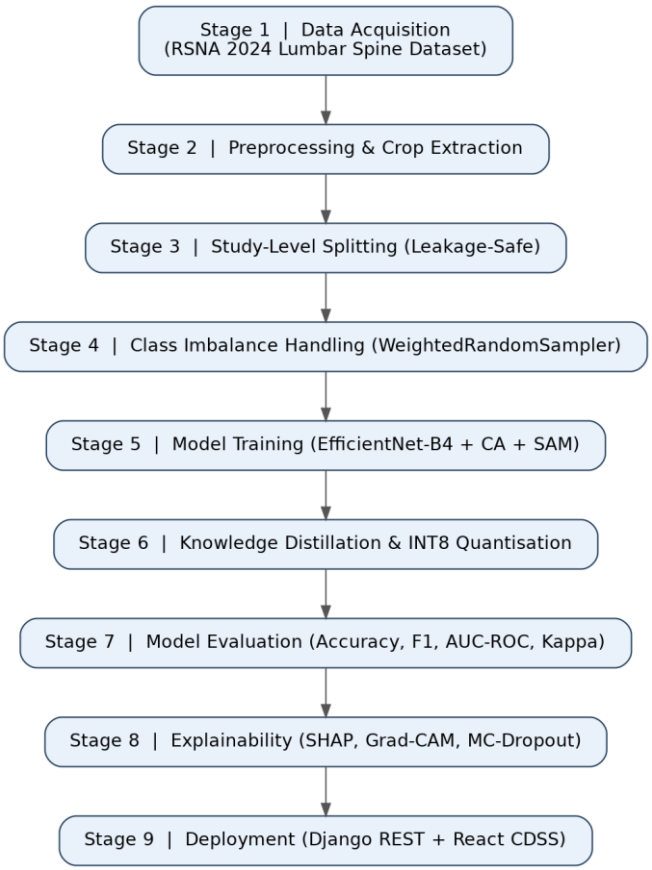


Figure 2: Methodology Flow Diagram – Spinova Pipeline

Figure 2 restates this pipeline as a simple nine-stage workflow, summarising the sequence in which each task was carried out, from data acquisition to deployment.

CHAPTER FOUR

FINDINGS AND DISCUSSION

4.1 Introduction

This chapter presents the results obtained from training and evaluating the Spinova classification model and the effect of model compression for edge deployment, states the functional and non-functional requirements derived for the resulting clinical decision support system, and presents the system design the use case diagram, entity-relationship diagram, and system flowchart that guided its development.

4.2 Data analysis results

The prepared crop-level dataset comprised 48,657 labelled crops in total, split into 29,269 training crops (1,186 studies), 9,746 validation crops (396 studies), and 9,642 test crops (392 studies), with no study leakage across splits. Table 1 summarises the class distribution of the test set used for final evaluation.

Table 1: Test-set class distribution (n = 9,642 crops from 392 studies)

Severity Class	Test Crops (n = 9,642)	Proportion
Normal/Mild	7 344	76%
Moderate	1 674	17%
Severe	624	6%

Figure 3 presents this same class distribution graphically.

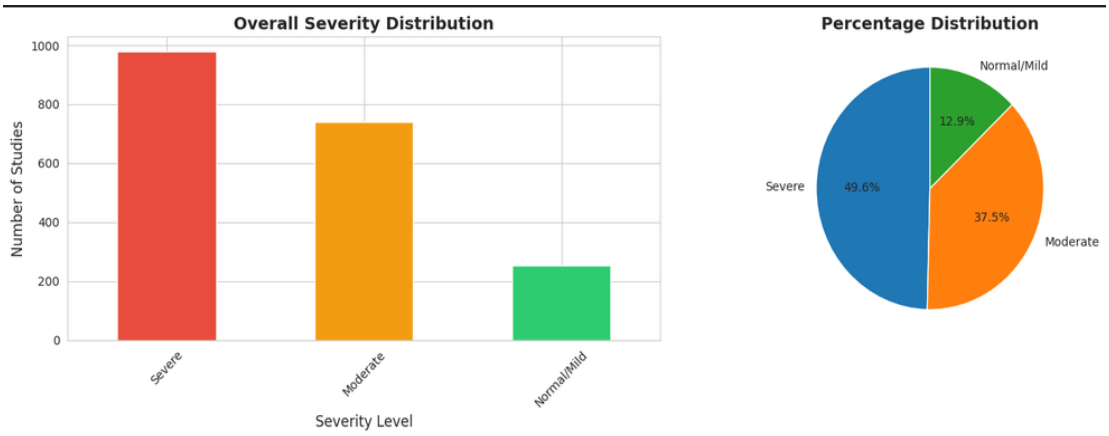


Figure 3: Test-Set Severity Class Distribution (Normal/Mild, Moderate, Severe)

Figure 3: Test-Set Severity Class Distribution (Normal/Mild, Moderate, Severe)

The hybrid EfficientNet-B4 + Coordinate Attention + Spatial Attention Module model achieved the crop-level test results summarised in Table 2.

Table 2: Hybrid model crop-level test performance

Metric	Result
Accuracy	79.15%
Precision (Macro)	0.599
Recall / Sensitivity (Macro)	0.628
F1-score (Macro)	0.594
AUC-ROC (Macro)	0.853
Cohen's Kappa	0.415
Weighted log loss	0.703

Per-class results showed strong performance on the majority Normal/Mild class (precision 0.86, recall 0.93, F1 0.89) but weaker performance on Moderate cases (precision 0.42, recall 0.22, F1 0.29) and moderate performance on Severe cases (precision 0.51, recall 0.74, F1 0.60). Post-hoc threshold tuning on the validation set selected a Severe-class decision threshold of 0.450, which produced the best achievable macro-F1 (0.593) without retraining. Test-Time Augmentation using five views raised overall accuracy to 80.48%, while Cohen's Kappa fell to 0.371 and Moderate-class recall dropped to 0.10, indicating that TTA improved majority-class agreement but further suppressed detection of the already under-represented Moderate class.

Figure 4 presents these per-class precision, recall, and F1-score results graphically, highlighting the gap between the strong Normal/Mild performance and the weaker, harder-to-separate Moderate class discussed above.

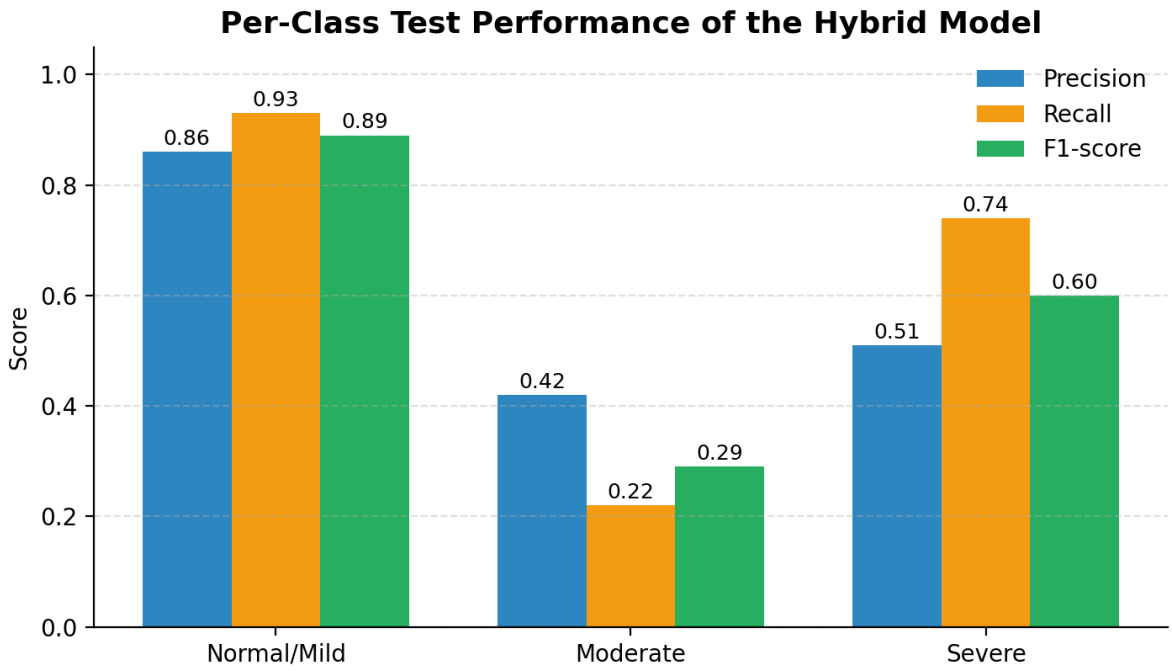


Figure 4: Per-Class Precision, Recall and F1-Score of the Hybrid Model

Figure 5 summarises the effect of distillation and INT8 quantisation on the deployed student model, showing the small reduction in model size achieved for essentially no loss in accuracy.

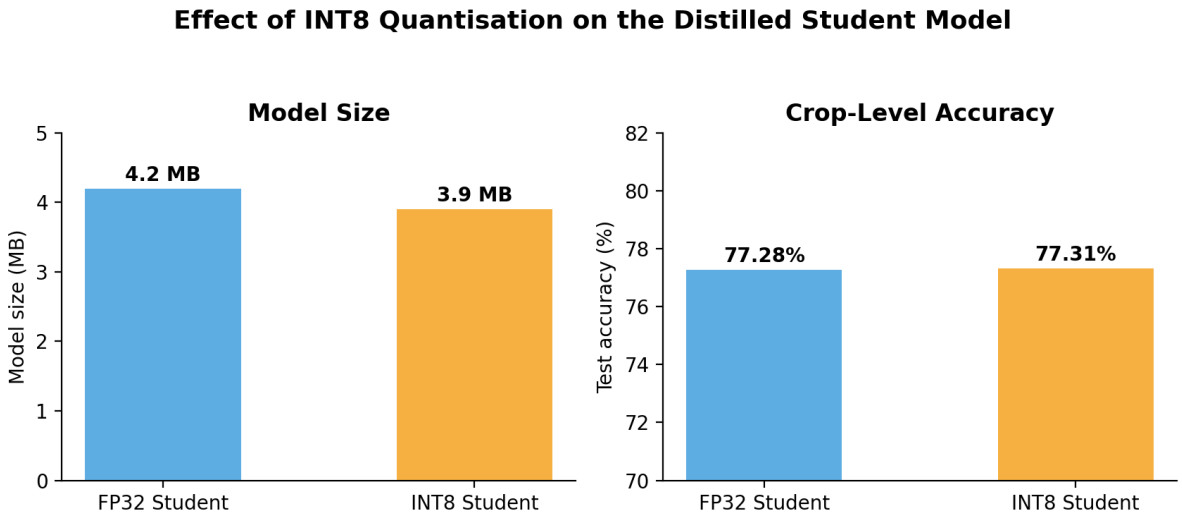


Figure 5: Effect of Distillation and INT8 Quantisation on Model Size and Accuracy

On explainability, SHAP attribution maps and Monte Carlo Dropout uncertainty estimates were successfully generated for individual predictions, providing two of the three planned explainability views without requiring a segmentation model; the third, mask-constrained Grad-CAM, remains dependent on a 3-D segmentation model that was identified as out of scope for this phase and is listed as future work in Chapter Five. The MRI Gatekeeper, built on a frozen MobileNetV3-Small embedding backbone with a Mahalanobis-style distance check against a reference distribution of known-good lumbar spine slices, was exported alongside its reference

statistics and reject threshold, giving the deployed system a functioning first line of defence against out-of-distribution or non-spine inputs.

Knowledge distillation into a MobileNetV3-Small student, followed by INT8 post-training quantisation, reduced model size from 4.2 MB (FP32) to 3.9 MB (INT8), a 5% reduction, while accuracy fell only marginally, from 77.28% (FP32 student) to 77.31% (INT8, within noise), a negligible −0.03 percentage-point accuracy delta attributable to quantisation. A simulated full-study inference (25 crops, covering five vertebral levels across five conditions) completed in 0.40 seconds on CPU, well within the project's under-10-second target for an 8GB-RAM hospital laptop. Both the quantised student model and the gatekeeper backbone were exported to ONNX format together with a model_config.json file specifying preprocessing constants and class names, and a standalone reload test confirmed that the exported ONNX artefacts function correctly without any dependency on the original training-time notebook state.

These results confirm the central hypothesis drawn from the literature reviewed in Chapter Two — that localized, crop-based classification with attention mechanisms (System 2) is better suited to this task than whole-slice classification (System 1): the model attends to a well-posed region rather than a mostly irrelevant background, and per-class results show the model performs strongly on the dominant class, while the Moderate class remains the hardest to separate from its neighbours. The gap between the achieved metrics and more ambitious targets (for example, 79.15% versus a target of 85% accuracy, and 0.415 versus 0.70 Kappa) indicates that architectural and attention improvements alone were insufficient to close that gap, and that further gains are more likely to come from richer training data, an ordinal-aware loss function, or an automatic landmark-localisation stage than from further threshold tuning on the current model. On the deployment side, the negligible accuracy loss from INT8 quantisation and the sub-second simulated inference time support the premise that an accurate-enough model can be made to run comfortably on CPU-only hospital hardware, which is central to the project's significance for resource-constrained settings such as Mbeya.

4.3 Functional Requirements

Table 3 lists the functional requirements derived for the Spinova system based on the objectives set out in Chapter One and the use cases described in Section 4.5.1.

Table 3: Functional requirements of the Spinova system

ID	Requirement

FR1	The system shall allow a clinician to register and sign in using authenticated credentials.
FR2	The system shall allow a clinician to upload a lumbar spine MRI study in DICOM format.
FR3	The system shall allow a clinician to mark level and condition points on the uploaded image.
FR4	The system shall reject images that the MRI Gatekeeper module identifies as out-of-distribution (non-lumbar-spine).
FR5	The system shall classify each marked point into Normal/Mild, Moderate, or Severe severity.
FR6	The system shall display the predicted severity, confidence score, and predictive uncertainty for each prediction.
FR7	The system shall generate and display SHAP and Grad-CAM explainability visualisations for a prediction.
FR8	The system shall map predicted severities to a surgical risk-tier flag for clinician review.
FR9	The system shall allow an administrator to manage registered users and deployed model artefacts.

4.4 Non-functional Requirements

Table 4 lists the non-functional requirements that constrain how the functional requirements above are to be met, reflecting the project's emphasis on deployability in resource-constrained clinical settings.

Table 4: Non-functional requirements of the Spinova system

Category	Requirement
Performance	The system shall complete inference for a full study (25 crops) in under 10 seconds on an 8GB-RAM, CPU-only hospital laptop.
Reliability	The system shall handle missing, corrupted, or unreadable DICOM files without crashing, returning a clear error instead.
Usability	The clinical workspace shall be operable by a clinician without machine-learning expertise, using point-and-click marking.
Security	All authenticated endpoints shall require a valid JSON Web Token (JWT) issued at sign-in.

Portability	The trained models shall run via ONNX Runtime without requiring a full PyTorch installation on the deployment server.
Maintainability	Preprocessing constants and class names shall be read from exported configuration files rather than hardcoded in source code.
Explainability	Every severity prediction shall be accompanied by at least one interpretability artefact rather than a bare class label.

4.5 System Design

This section presents the design artefacts produced for the Spinova system: an overview of its layered architecture, the use case diagram describing how actors interact with the system, the entity-relationship diagram describing its data model, and a flowchart describing the end-to-end classification workflow.

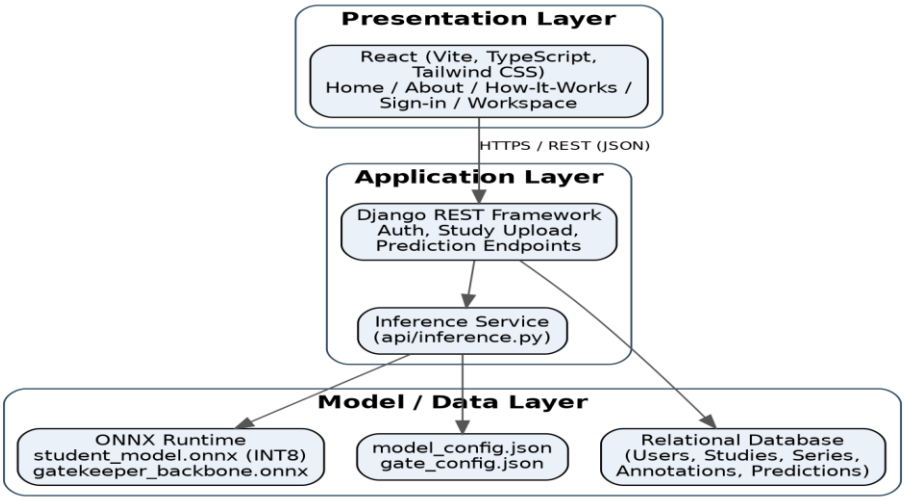


Figure 6: Spinova System Architecture

As shown in Figure 6, Spinova is organised into three layers: a presentation layer (the React frontend, providing the clinical workspace and marketing pages), an application layer (the Django REST Framework backend, handling authentication, study upload, and prediction endpoints), and a model/data layer (the ONNX Runtime inference service, exported model artefacts and configuration files, and the relational database storing users, studies, series, annotations, and predictions). This layering keeps the machine-learning artefacts decoupled from the web application logic, so that the model can be retrained and re-exported without requiring changes to the backend or frontend code.

4.5.1 Use Case

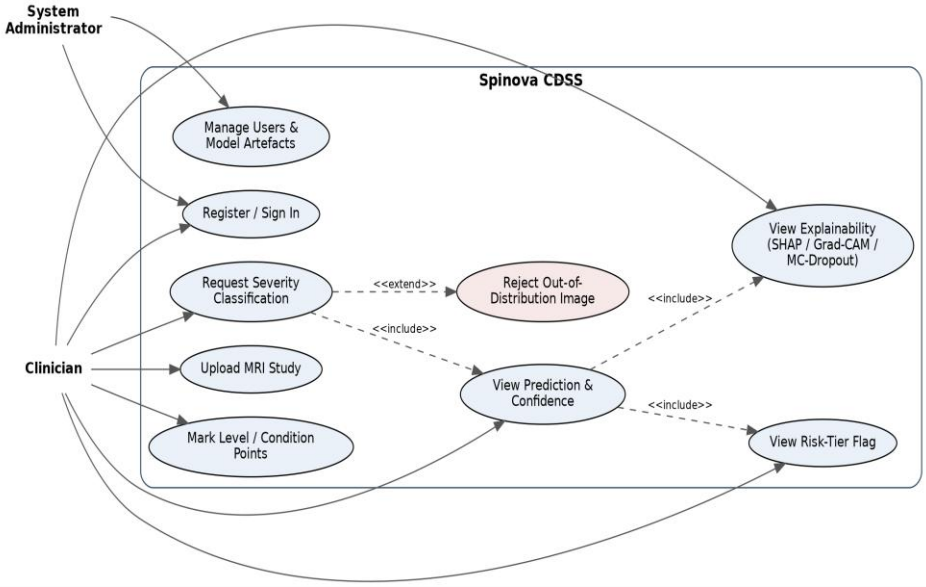


Figure 7: Use Case Diagram for the Spinova CDSS

Figure 7 shows the two actors who interact with the Spinova system — the Clinician and the System Administrator — and the use cases available to each. A Clinician registers or signs in, uploads an MRI study, marks level and condition points on the image, and requests a severity classification. The "Request Severity Classification" use case includes viewing the resulting prediction and confidence, which in turn includes viewing the explainability outputs (SHAP, Grad-CAM, and MC-Dropout uncertainty) and the surgical risk-tier flag, and is extended by the "Reject Out-of-Distribution Image" use case whenever the MRI Gatekeeper determines that the uploaded image is not a lumbar spine MRI. The System Administrator is responsible for managing registered users and the deployed model artefacts.

4.5.2 E-R diagram

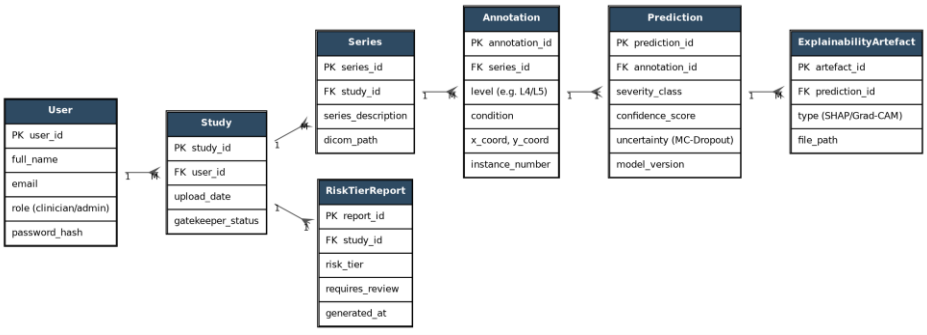


Figure 8:Entity-Relationship Diagram for the Spinova Database

Figure 8 shows the entity-relationship diagram underlying the Spinova database. A User (clinician or administrator) owns many Studies, each of which contains many Series (individual DICOM series within an MRI study). Each Series carries many Annotations, corresponding to the level/condition points marked by the clinician; each Annotation produces exactly one Prediction from the classification model, and each Prediction may have several associated

ExplainabilityArtefacts (SHAP maps, Grad-CAM overlays). Each Study is also associated with one RiskTierReport, which aggregates the study's predictions into a single surgical risk-tier flag for clinician review.

4.5.3 Flowchart

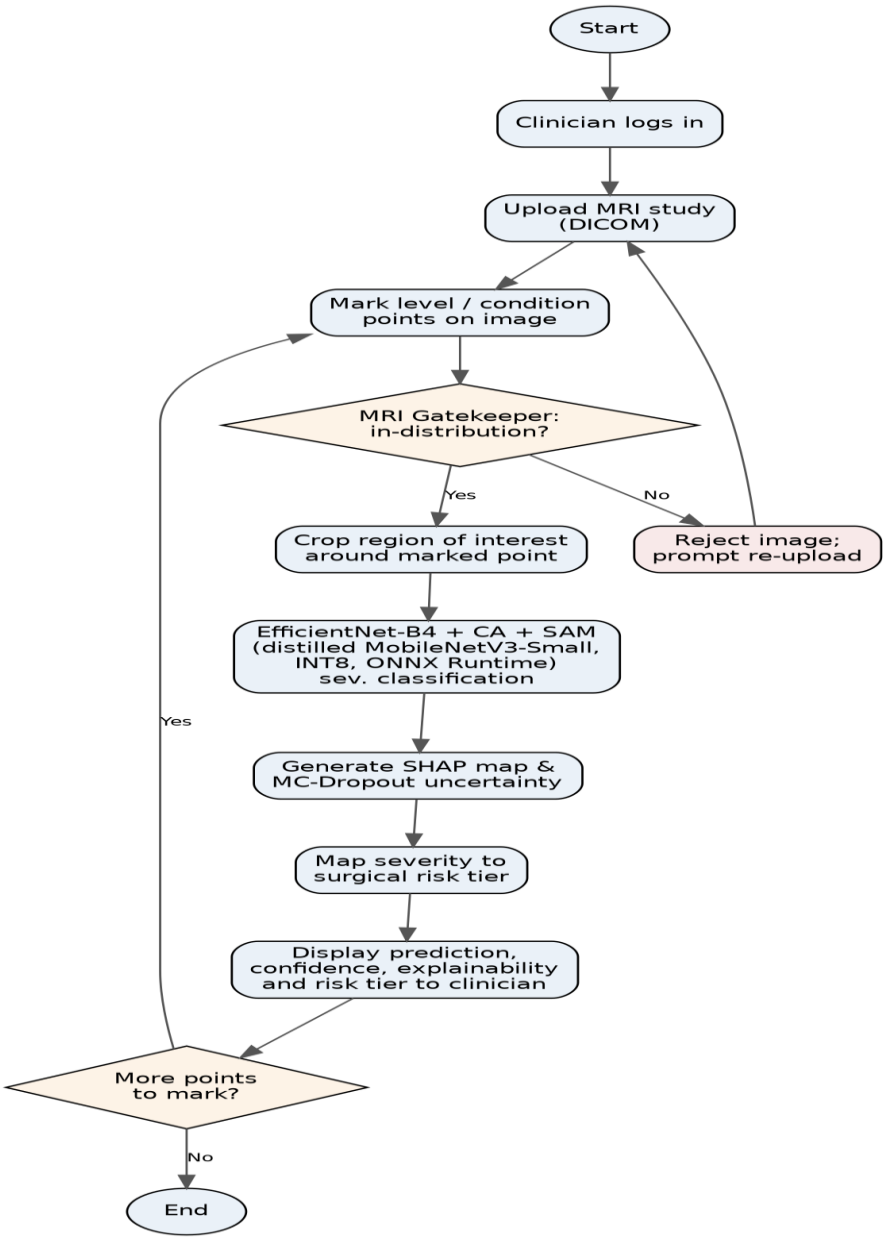


Figure 9: System Flowchart for the Spinova Classification Workflow

Figure 9 traces the end-to-end workflow of a single classification request. After signing in, a clinician uploads an MRI study and marks a level/condition point on the image. The MRI Gatekeeper first checks whether the image is in-distribution; if not, the image is rejected and the clinician is prompted to re-upload. If the image passes the gatekeeper check, the region of interest around the marked point is cropped and passed to the distilled, INT8-quantised classification model running through ONNX Runtime, which returns a severity prediction. SHAP and MC-Dropout outputs are then generated, the predicted severity is mapped to a

surgical risk tier, and the full result prediction, confidence, explainability, and risk tier is displayed to the clinician, who may then mark further points on the same study or end the session.

CHAPTER FIVE

CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

This project set out to design, develop, and evaluate Spinova, a deep learning based Clinical Decision Support System for classifying the severity of lumbar spine degenerative conditions from MRI. A two-stage, localize-then-classify pipeline was built around an EfficientNet-B4 backbone with Coordinate Attention and a Spatial Attention Module, trained on the RSNA 2024 Lumbar Spine Degenerative Classification dataset using study-level stratified splits, class-weighted loss, oversampling, and augmentation to manage class imbalance. An MRI Gatekeeper module for out-of-distribution rejection, SHAP and Monte Carlo Dropout explainability, knowledge distillation to a MobileNetV3-Small student, INT8 quantisation, and ONNX export were added to make the system explainable and deployable. The compressed model was integrated into a Django REST Framework backend and a React frontend, forming a working clinical workspace for image upload, point-based marking, and result review.

The project concludes that a localized, attention-guided classification approach is a sound and literature-consistent design choice for lumbar spine MRI severity grading, achieving a crop-level test accuracy of 79.15% and an AUC-ROC of 0.853, though it fell short of more ambitious targets, particularly on the Moderate severity class. The project further concludes that model compression through distillation and INT8 quantisation can be achieved with negligible accuracy loss, making CPU-only, sub-second full-study inference realistic for resource-constrained clinical settings, and that explainability and out-of-distribution safeguards can be partially integrated into such a pipeline without a segmentation model, though a fully complete explainability suite still depends on 3-D segmentation work not undertaken in this project. Overall, the objectives of the project were substantially achieved, and a functioning end-to-end prototype, from trained model to clinical workspace, was delivered.

5.2 Recommendations

i. The classification model should be further improved by exploring ordinal-aware loss functions and additional training data or augmentation strategies targeted specifically at the Moderate severity class, which showed the weakest performance.

ii. An automatic vertebral-level and landmark localisation model should be developed so that the system can generate its own region-of-interest coordinates on new, unlabelled clinical scans,

since the current pipeline depends on ground-truth coordinates that will not exist for real patients.

iii. A 3-D segmentation model should be developed to enable mask-constrained Grad-CAM, completing the planned three-view explainability suite alongside the SHAP and Monte Carlo Dropout views already implemented.

iv. Before any clinical use, the surgical risk-tier thresholds currently used in the system should be reviewed and validated by a qualified clinician or radiologist, and the system as a whole should undergo prospective clinical validation and appropriate regulatory review.

v. Areas for further research include extending the pipeline to other spinal regions, evaluating the system prospectively in a real Tanzanian clinical setting, and investigating federated or privacy-preserving training approaches that would allow the model to be improved using data from multiple hospitals without centralising sensitive patient images.

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APPENDICES

Appendix I: Deployed Model Configuration (model_config.json)

IMG_SIZE: 224; CROP_FRAC: 0.18; Normalisation: ImageNet mean [0.485, 0.456, 0.406], std [0.229, 0.224, 0.225]; CLASS_NAMES: Normal/Mild, Moderate, Severe; CONDITIONS: Spinal Canal Stenosis, Left Neural Foraminal Narrowing, Right Neural Foraminal Narrowing, Left Subarticular Stenosis, Right Subarticular Stenosis; MODEL_VERSION: LSS_Student-int8-v0.1.

Appendix II: System Route Map (React Frontend) and Exported Model Artefacts

Route map / : Home page (hero and 3-step teaser); /about: disease information page; /how-it-works: walkthrough of the screening flow; /signin: sign-in / registration; /workspace: authenticated study-review dashboard for uploading images, marking points, and viewing predictions.

Exported model artefacts (Django backend/ml_artifacts/) student_model.onnx (quantised INT8 severity classifier); gatekeeper_backbone.onnx (MobileNetV3-Small embedding backbone); model_config.json (preprocessing constants and class names); gate_config.json (reference statistics and out-of-distribution reject threshold).